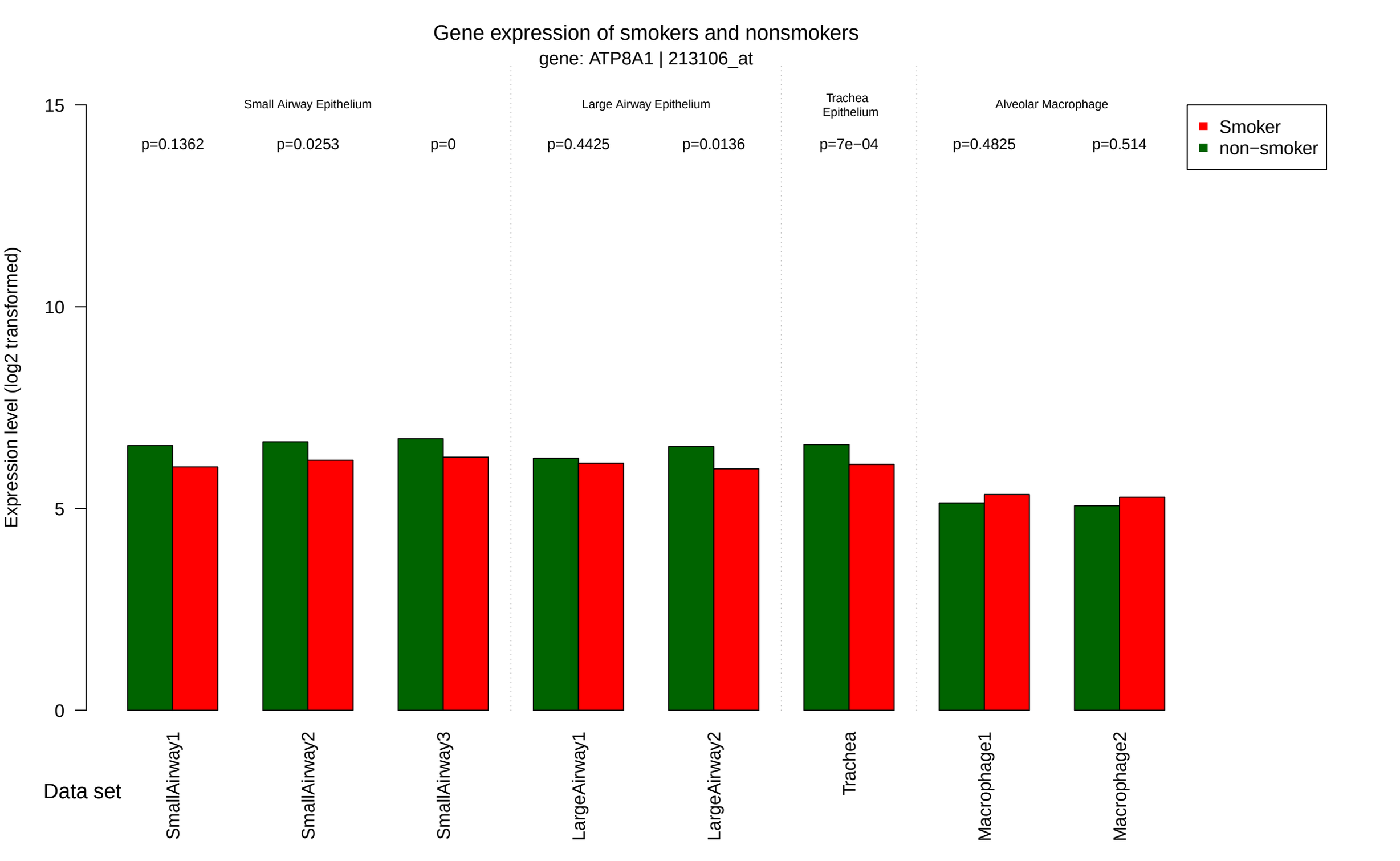
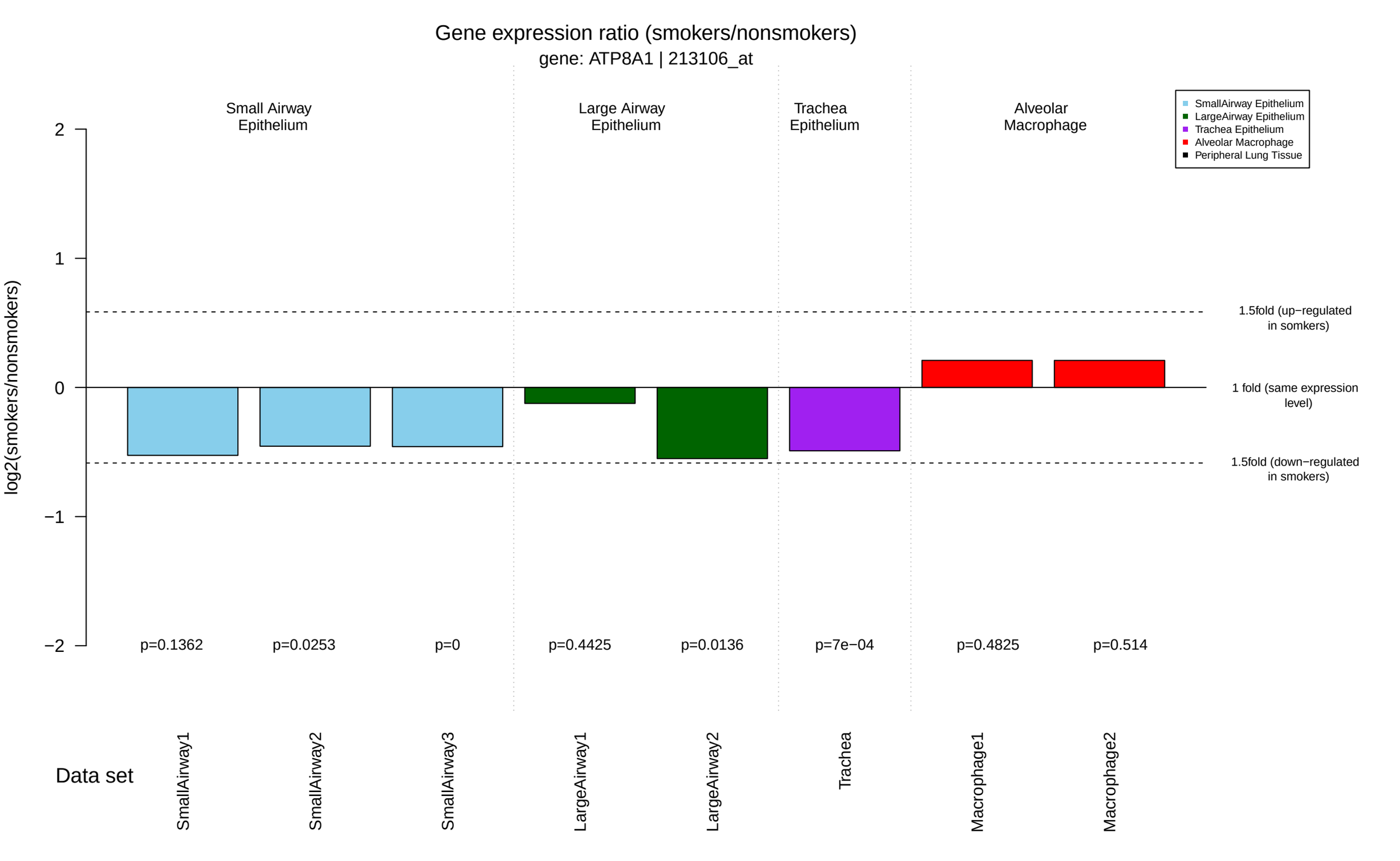
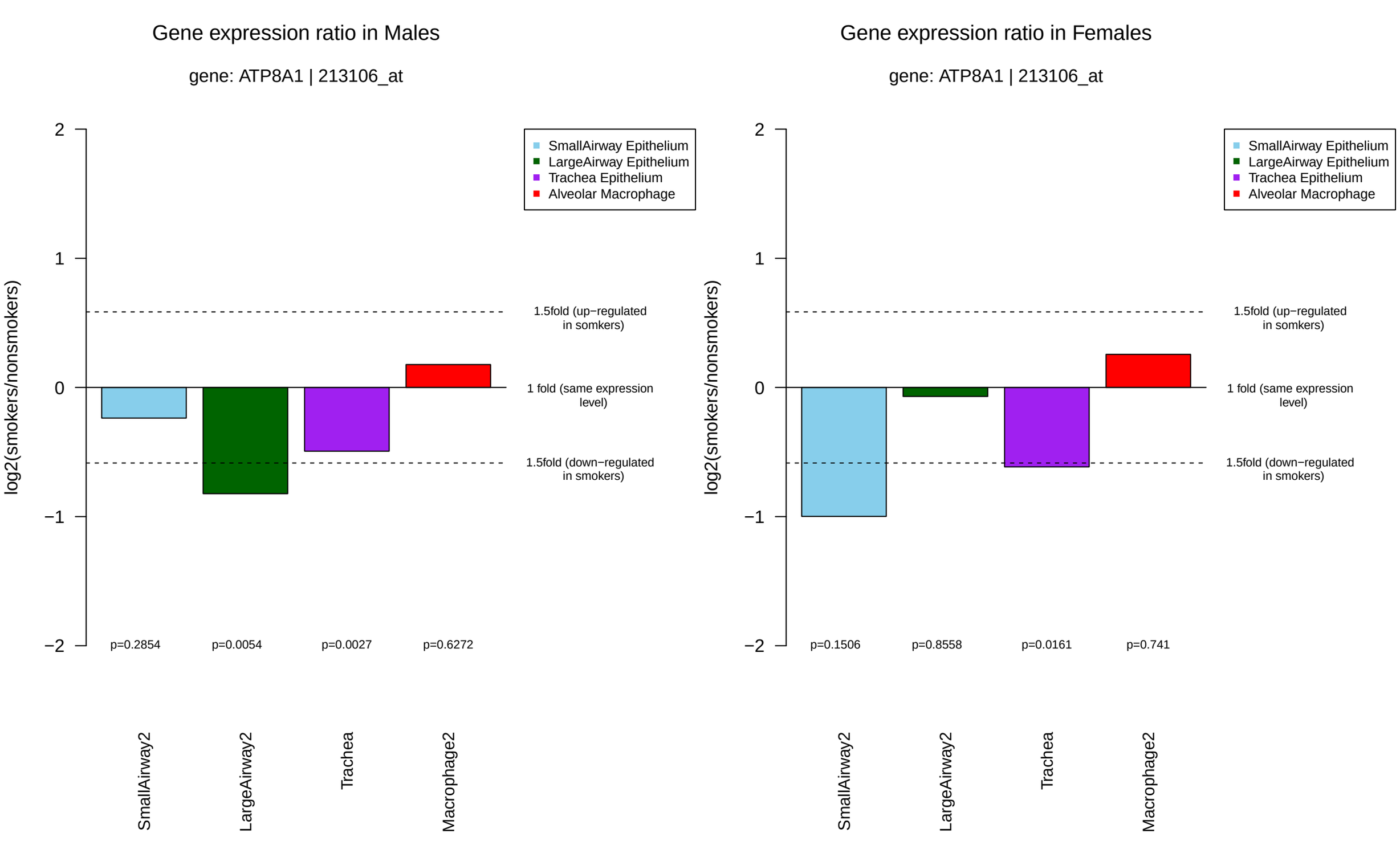
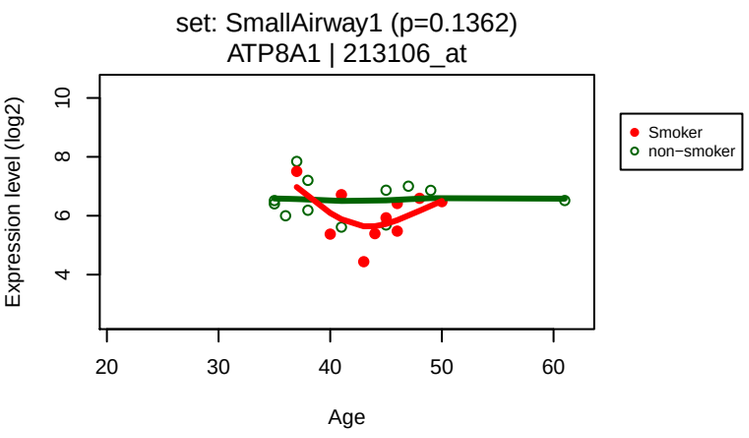
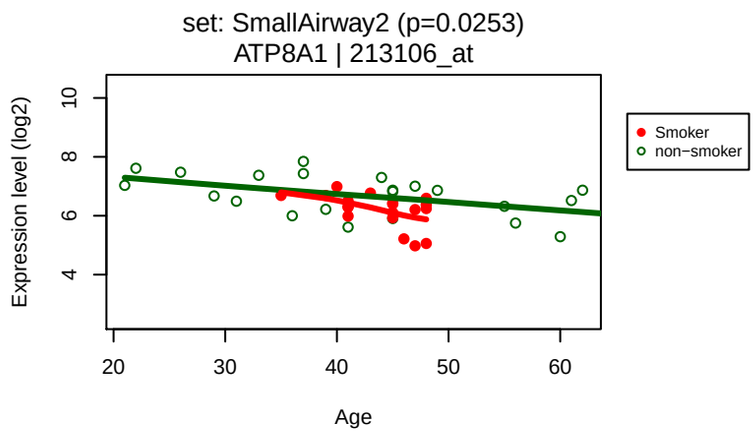
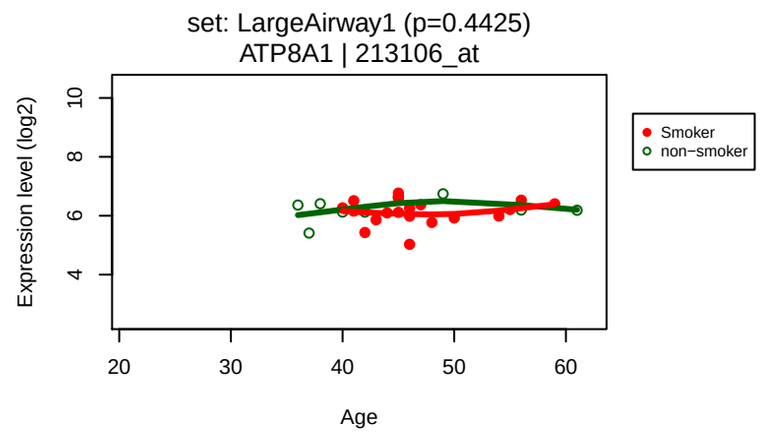

Gene expression of smokers and nonsmokers
gene: ATP8A1 | 213106_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1362
p=0.0253
p=0
p=0.4425
p=0.0136
p=7e−04
p=0.4825
p=0.514
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

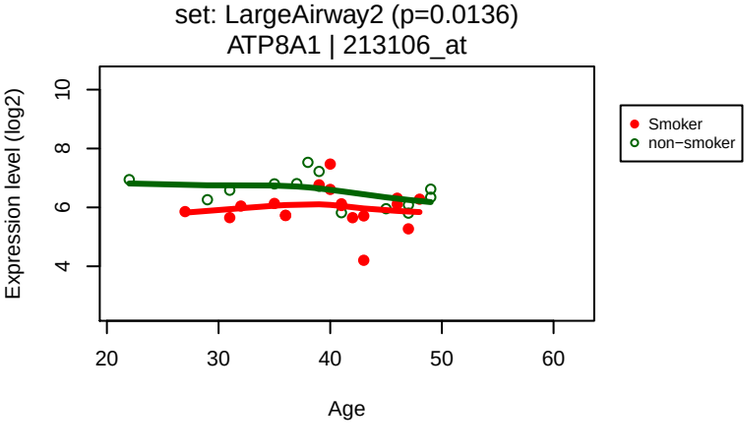

Gene expression ratio (smokers/nonsmokers)
gene: ATP8A1 | 213106_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1362
p=0.0253
p=0
p=0.4425
p=0.0136
p=7e−04
p=0.4825
p=0.514
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

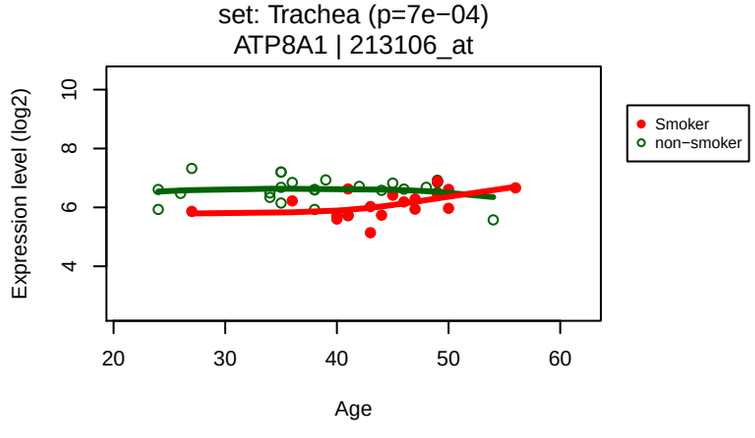

Gene expression ratio in Males
gene: ATP8A1 | 213106_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2854
p=0.0054
p=0.0027
p=0.6272
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP8A1 | 213106_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1506
p=0.8558
p=0.0161
p=0.741
SmallAirway2
LargeAirway2
Trachea
Macrophage2

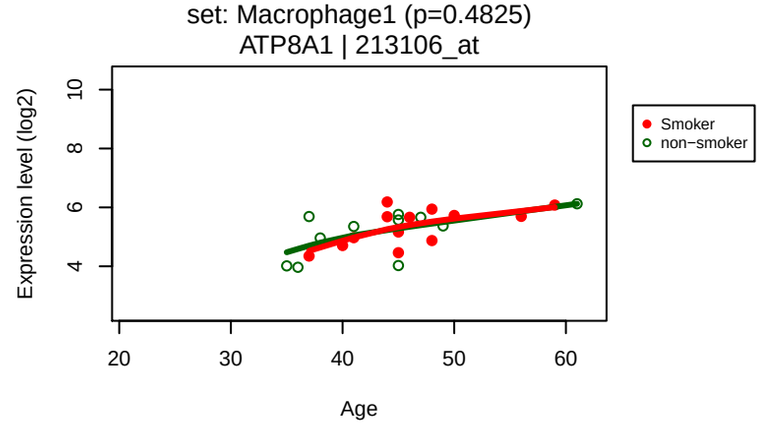

set: SmallAirway1 (p=0.1362)
ATP8A1 | 213106_at
Expression level (log2)
Age
Smoker
non−smoker

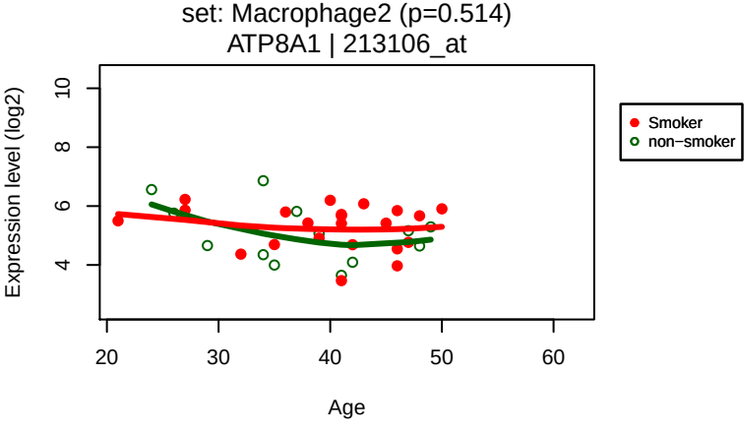

set: SmallAirway2 (p=0.0253)
ATP8A1 | 213106_at
Expression level (log2)
Age
Smoker
non−smoker

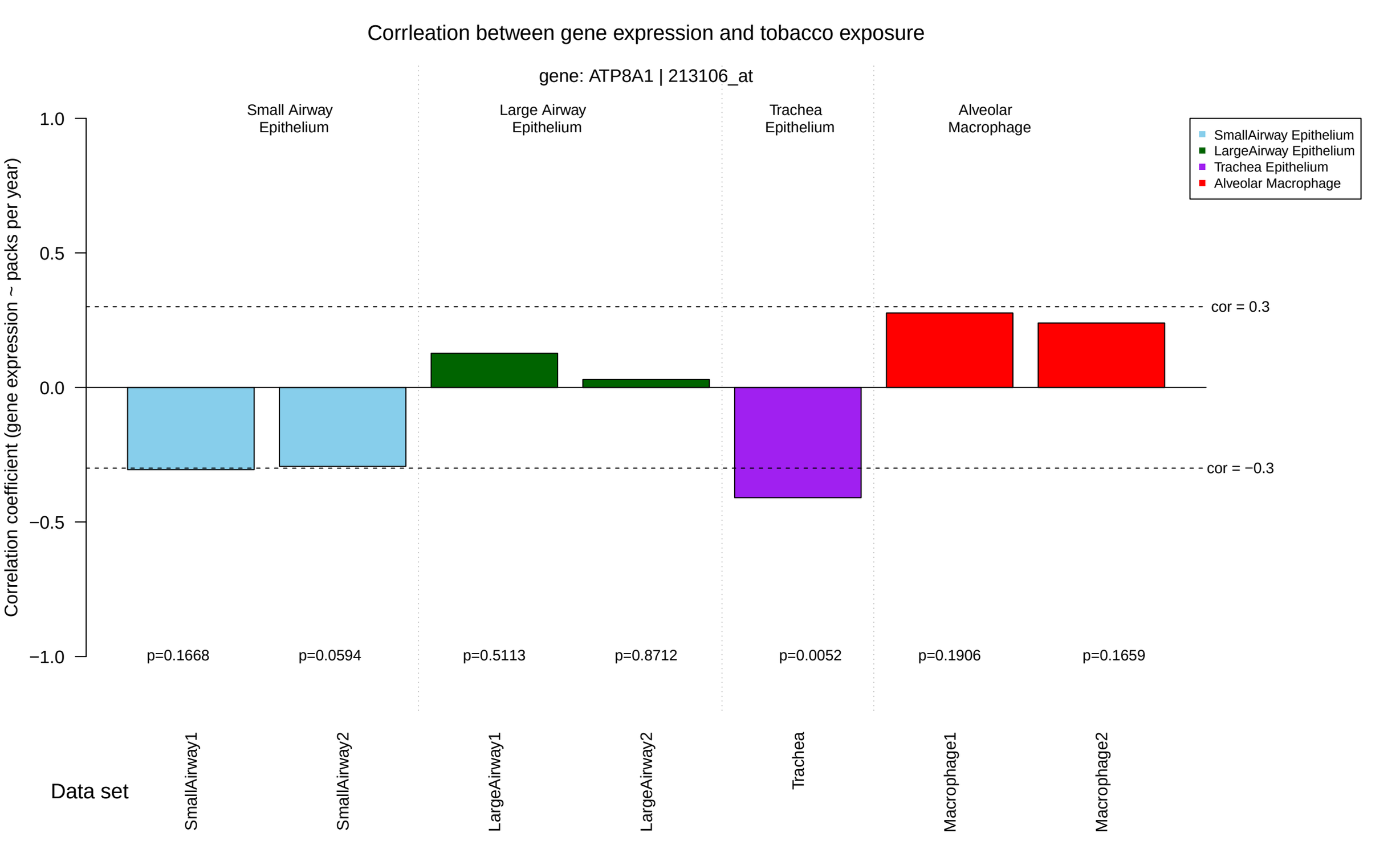

set: LargeAirway1 (p=0.4425)
ATP8A1 | 213106_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0136)
ATP8A1 | 213106_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=7e−04)
ATP8A1 | 213106_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4825)
ATP8A1 | 213106_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.514)
ATP8A1 | 213106_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP8A1 | 213106_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1668
p=0.0594
p=0.5113
p=0.8712
p=0.0052
p=0.1906
p=0.1659
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2